```c
#include<stdio.h>

int main()
{
    int a[5]={10,20,30,40,50};
    int n,low=0,high=4,mid;

    printf("Enter roll number: ");
    scanf("%d",&n);

    while(low<=high)
    {
        mid=(low+high)/2;

        if(a[mid]==n)
        {
            printf("Roll number found");
            return 0;
        }
        else if(a[mid]<n)
            low=mid+1;
        else
            high=mid-1;
    }

    printf("Roll number not found");

    return 0;
}
```

main.c

Clear Run ▶

```c
#include<stdio.h>

int main()
{
    int a[5]={10,20,30,40,50};
    int n,low=0,high=4,mid;

    printf("Enter roll number: ");
    scanf("%d",&n);

    while(low<=high)
    {
        mid=(low+high)/2;

        if(a[mid]==n)
        {
            printf("Roll number found");
            return 0;
        }
        else if(a[mid]<n)
            low=mid+1;
        else
```

```c
while(low<=high)
{
    mid=(low+high)/2;

    if(a[mid]==n)
    {
        printf("Roll number found");
        return 0;
    }
    else if(a[mid]<n)
        low=mid+1;
    else
        high=mid-1;
}

printf("Roll number not found");

return 0;
```

Output

```
Enter roll number: 1
Roll number not found
```